Huawei Access Node BTM-ACC-34 Optical Transport Failure SOP

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_2626

Description

This SOP outlines the procedures for troubleshooting and resolving an Optical Transport Failure on the Huawei Access Node BTM-ACC-34 at the BTM Layout Aggregation site. The failure domain is attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Optical Transport Network (OTN) equipment on the Huawei Access Node operates on the OTU2/OTU3 protocol stack, with the OTU2/OTU3 framing and FEC (Forward Error Correction) enabled. The OTN equipment uses the WDM (Wavelength Division Multiplexing) technology for transmitting multiple channels over a single fiber. The failure is likely caused by the degradation of the SFP/XFP, which houses the optical transceiver, or a fiber cut, which disrupts the light signal transmission. The Rx/Tx power loss can be caused by a combination of factors, including a faulty SFP/XFP, a dirty or damaged fiber, or a misconfigured OTN equipment.

Pre-requisite Checks

- Check: Verify the site location and confirm that the issue is isolated to the Huawei Access Node BTM-ACC-34 at the BTM Layout Aggregation site.

- Check: Check the redundant link status to ensure that the failure has not affected the backup link.

- Check: Dispatch remote hands to the BTM Layout Aggregation site to physically inspect the fiber connections and SFP/XFP modules.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display opt trans information, Expected Output: Optical transmission information, Description: This command displays the optical transmission information, including the SFP/XFP status, Rx/Tx power levels, and OTU2/OTU3 framing and FEC status.

Command: display opt trans error, Expected Output: Optical transmission error information, Description: This command displays the optical transmission error information, including the error counts, error rates, and alarm states.
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Command: display opt trans alarm, Expected Output: Optical transmission alarm information, Description: This command displays the optical transmission alarm information, including the alarm codes, severity levels, and description.
```

Resolution Steps

Step 1: Step: Check SFP/XFP status, Command: display opt trans information | include SFP, Description: Check the SFP/XFP status to determine if it is faulty or degraded. If the SFP/XFP is faulty or degraded, replace it with a new one.

Step 2: Step: Check fiber connections, Command: display opt trans information | include fiber, Description: Check the fiber connections to ensure that they are properly connected and not damaged. If the fiber connections are damaged, replace them with new ones.

Step 3: Step: Check OTU2/OTU3 framing and FEC, Command: display opt trans information | include OTU2, Description: Check the OTU2/OTU3 framing and FEC status to ensure that they are properly configured. If the OTU2/OTU3 framing and FEC are misconfigured, reconfigure them to the correct settings.

Post-Check Verification

```
Command: display opt trans information, Expected Output: Optical transmission information, Description: Verify that the optical transmission information displays the correct SFP/XFP status, Rx/Tx power levels, and OTU2/OTU3 framing and FEC status.

Command: display opt trans error, Expected Output: Optical transmission error information, Description: Verify that the optical transmission error information displays the correct error counts, error rates, and alarm states.

Command: display opt trans alarm, Expected Output: Optical transmission alarm information, Description: Verify that the optical transmission alarm information displays the correct alarm codes, severity levels, and description.
```

Roll-back Procedures

- Step: Restore original configuration, Command: undo opt trans configuration, Description: Restore the original configuration to ensure that the system returns to its previous state.

- Step: Reboot the device, Command: reboot, Description: Reboot the device to ensure that the changes are fully applied.